

Step 1: Create folder project

```
nguyet@ltn:~$ cd Documents/  
nguyet@ltn:~/Documents$ mkdir my-openwrt-checkpy  
nguyet@ltn:~/Documents$ cd my-openwrt-checkpy/  
nguyet@ltn:~/Documents/my-openwrt-checkpy$ mkdir checkpy  
nguyet@ltn:~/Documents/my-openwrt-checkpy$ cd checkpy/  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy$ touch Makefile  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy$ mkdir src  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy$ cd src  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy/src$ touch checkpy.c  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy/src$ nano checkpy.c  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy/src$ cd ..  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy$ nano Makefile  
nguyet@ltn:~/Documents/my-openwrt-checkpy/checkpy$ cd ..  
nguyet@ltn:~/Documents/my-openwrt-checkpy$ tree
```

```
.  
├── checkpy  
│   ├── Makefile  
│   └── src  
│       └── checkpy.c  
  
2 directories, 2 files
```

Step 2: Write Dockerfile, Makefile and checkpy.c

Step 3: Build image

```
nguyet@ltn:~/Documents/my-openwrt-checkpy$ docker build -t openwrt-builder .  
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.  
Install the buildx component to build images with BuildKit:  
https://docs.docker.com/go/buildx/  
  
Sending build context to Docker daemon 6.144kB  
Step 1/10 : FROM ubuntu:22.04  
--> 1d3ca894b30c  
Step 2/10 : ENV DEBIAN_FRONTEND=noninteractive  
--> Using cache  
--> 5699d1b0f699  
Step 3/10 : RUN apt-get update && apt-get install -y sngrep yate  
--> 7d6380ef7734  
Step 4/10 : RUN apt-get update && apt-get install -y sngrep yate  
--> 7d6380ef7734  
Step 5/10 : RUN apt-get update && apt-get install -y sngrep yate  
--> 7d6380ef7734  
Step 6/10 : RUN apt-get update && apt-get install -y sngrep yate  
--> 7d6380ef7734  
Step 7/10 : RUN apt-get update && apt-get install -y sngrep yate  
--> 7d6380ef7734  
Step 8/10 : RUN apt-get update && apt-get install -y sngrep yate  
--> 7d6380ef7734  
Step 9/10 : RUN apt-get update && apt-get install -y sngrep yate  
--> 7d6380ef7734  
Step 10/10 : CMD ["/bin/bash"]  
--> Running in 3399e22f8d9e  
--> Removed intermediate container 3399e22f8d9e  
--> 328c6f2c8fad  
Successfully built 328c6f2c8fad  
Successfully tagged openwrt-builder:latest
```

Step 4: Run container with mount volume

Make menuconfig

```
nguyet@ltn:~/Documents/my-openwrt-checkpy$ docker run -it \  
-v $(pwd)/checkpy:/home/builder/openwrt/package/checkpy \  
--name openwrt-dev \  
openwrt-builder  
builder@f5a3fda49dbd:~/openwrt$ cd /home/builder/openwrt  
builder@f5a3fda49dbd:~/openwrt$ make menuconfig
```

Target System

Use the arrow keys to navigate this window or press the hotkey of the item you wish to select followed by the <SPACE BAR>. Press <?> for additional information about this

- ^(-)
- () Allwinner A1x/A20/A3x/H3/H5/R40
 - () AppliedMicro APM821xx
 - () Arm SystemReady (EFI) compliant
 - () Atheros AR231x/AR5312
 - () Atheros ATH79
 - (X) **Broadcom BCM27xx**

v(+)

<Select>

< Help >

Subtarget

Use the arrow keys to navigate this window or press the hotkey of the item you wish to select followed by the <SPACE BAR>. Press <?> for additional information about this

- () BCM2708 boards (32 bit)
- () BCM2709/BCM2710/BCM2711 boards (32 bit)
- () BCM2710 boards (64 bit)
- (X) **BCM2711 boards (64 bit)**

<Select>

< Help >

Target Profile

Use the arrow keys to navigate this window or press the hotkey of the item you wish to select followed by the <SPACE BAR>. Press <?> for additional information about this

- () Multiple devices
- (X) **Raspberry Pi 4B/400/CM4 (64bit)**

<Select>

< Help >

```

Administration --->
Boot Loaders --->
Development --->
Extra packages --->
Firmware --->
Fonts --->
Kernel modules --->
Languages --->
Libraries --->
LuCI --->
Mail --->
Multimedia --->
Network --->
Sound --->
Utilities --->
Xorg --->

```

```

^(-)
< > bossa..... Utility to flash ARM microcontrollers
< > bottlerocket..... Bottlerocket
< > bsdiff..... Binary diff tool
< > bspatch..... Binary patch tool
< > byobu..... Text-based window manager and terminal multiplexer
< > byobu-utils..... Utilities included with byobu
< > cache-domains-mbedtls..... Game content cache DNS (mbedtls)
< > cache-domains-openssl..... Game content cache DNS (openssl)
< > cache-domains-wolfssl..... Game content cache DNS (wolfssl)
< > cal..... display a calendar
< > canutils..... CAN userspace utilities and tools ----
< > catatonit
< > cgroup-tools..... CGroup config and exec tools
< > cgroupfs-mount..... cgroup mount scripts ----
< > checkpolicy..... SELinux policy compiler
<*> checkpy..... Python 3.9 Checker
< > checksec..... Utility to check PIE, RELRO, ASLR etc.
< > checksec_automator..... Utility to use checksec for dirs
< > chkcon..... libsepol chkcon security context validation tool
< > clocate..... Determine device location using neighbouring WiFi networks
< > cmdpad..... execute commands when key is pressed/released/held down
< > cni..... cni
< > cni-plugins..... cni-plugins
< > cni-plugins-nft..... CNI Plugins compatible with nftables
< > coap-client..... CoAP (RFC 7252) client tool
< > collectd..... Lightweight system statistics collection daemon ----
< > colrm..... colrm removes selected columns from a file
< > common..... Podman common
< > containerd..... containerd container runtime ----
< > coremark..... CoreMark Embedded Microprocessor Benchmark
< > coreutils..... The GNU core utilities ----
v(+)

```

Step 5: Make package

```
builder@f5a3fda49dbd:~/openwrt$ make package/checkpy/compile V=s
make[2]: Entering directory '/home/builder/openwrt/scripts/config'
make[2]: 'conf' is up to date.
make[2]: Leaving directory '/home/builder/openwrt/scripts/config'
make[1]: Entering directory '/home/builder/openwrt'
+ mkdir -p /home/builder/openwrt/staging_dir/target-aarch64_cortex-a72_musl
+ cd /home/builder/openwrt/staging_dir/target-aarch64_cortex-a72_musl
+ mkdir -p bin lib stamp usr/include usr/lib
mkdir -p /home/builder/openwrt/build_dir/target-aarch64_cortex-a72_musl/stamp
touch /home/builder/openwrt/staging_dir/target-aarch64_cortex-a72_musl/.prepared
make[2]: Entering directory '/home/builder/openwrt/package/libs/toolchain'
echo "libc" >> /home/builder/openwrt/staging_dir/target-aarch64_cortex-a72_musl/pkginfo/toolchain.default.install
```

Step 6: Check file .ipk

```
builder@f5a3fda49dbd:~/openwrt$ ls /home/builder/openwrt/bin/packages/aarch64_cortex-a72/base/checkpy_1.0-1_aarch64_cortex-a72.ipk
/home/builder/openwrt/bin/packages/aarch64_cortex-a72/base/checkpy_1.0-1_aarch64_cortex-a72.ipk
```

Step 7: Extract file .ipk from host and copy file bin to Pi 4

```
nguyet@ltn:~$ mkdir ~/checkpy_extract
nguyet@ltn:~$ cd ~/checkpy_extract
nguyet@ltn:~/checkpy_extract$ cp ~/checkpy_1.0-1_aarch64_cortex-a72.ipk .
nguyet@ltn:~/checkpy_extract$ file checkpy_1.0-1_aarch64_cortex-a72.ipk
checkpy_1.0-1_aarch64_cortex-a72.ipk: gzip compressed data, from Unix, original size modulo 2^32 10240
nguyet@ltn:~/checkpy_extract$ mv checkpy_1.0-1_aarch64_cortex-a72.ipk checkpy.gz
nguyet@ltn:~/checkpy_extract$ file checkpy
checkpy: POSIX tar archive (GNU)
nguyet@ltn:~/checkpy_extract$ scp checkpy root@192.168.1.1:/tmp
scp: /tmp/checkpy: Is a directory
nguyet@ltn:~/checkpy_extract$ ls
checkpy
nguyet@ltn:~/checkpy_extract$ cd checkpy
bash: cd: checkpy: Not a directory
nguyet@ltn:~/checkpy_extract$ tar -xvf checkpy
./debian-binary
./data.tar.gz
./control.tar.gz
nguyet@ltn:~/checkpy_extract$ mkdir data && tar -xvf data.tar.gz -C data
./
./usr/
./usr/bin/
./usr/bin/checkpy
nguyet@ltn:~/checkpy_extract$ cd data
nguyet@ltn:~/checkpy_extract/data$ file ./usr/bin/checkpy
./usr/bin/checkpy: ELF 64-bit LSB executable, ARM aarch64, version 1 (SYSV), dynamically linked, interpreter /lib/ld-musl-aarch64.so.1, no section header
nguyet@ltn:~/checkpy_extract/data$ scp ./usr/bin/checkpy root@192.168.1.1:/tmp/
scp: /tmp/checkpy: Is a directory
nguyet@ltn:~/checkpy_extract/data$ scp ./usr/bin/checkpy root@192.168.1.1:/tmp/checkpy_bin
checkpy                               100% 64KB 8.2MB/s 00:00
nguyet@ltn:~/checkpy_extract/data$
```

Step 8: Run package in Pi4

```
root@openwrt:/tmp# /tmp/checkpy_bin
Detected Python Version: sh: python3.9: not found
root@openwrt:/tmp# python3 --version
-bash: python3: not found
root@openwrt:/tmp# opkg -h
-bash: opkg: not found
root@openwrt:/tmp# cd /
root@openwrt:/# ls
bin      lib      overlay  run      usr
root     lib64    proc     sbin     var
```

Step 9: Git commit

```
nguyet@ltn:~/Documents/my-openwrt-checkpy$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/nguyet/Documents/my-openwrt-checkpy/.git/
nguyet@ltn:~/Documents/my-openwrt-checkpy$ git checkout -b feature/python-version-check
Switched to a new branch 'feature/python-version-check'
nguyet@ltn:~/Documents/my-openwrt-checkpy$ git add .
nguyet@ltn:~/Documents/my-openwrt-checkpy$ git commit -m "Initial Python 3.9 check up"
[feature/python-version-check (root-commit) 2d1f8fa] Initial Python 3.9 check up
5 files changed, 126 insertions(+)
create mode 100644 Dockerfile
create mode 100644 Makefile
create mode 100644 README.md
create mode 100644 checkpy/Makefile
create mode 100644 checkpy/src/checkpy.c
```

```
and the repository entered
nguyet@ltn:~/Documents/my-openwrt-checkpy$ git remote add origin https://github.com/virgogoo/python-check.git
nguyet@ltn:~/Documents/my-openwrt-checkpy$ git push origin feature/python-version-check
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 1.93 KiB | 990.00 KiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/virgogoo/python-check.git
 * [new branch]      feature/python-version-check -> feature/python-version-check
nguyet@ltn:~/Documents/my-openwrt-checkpy$ git remote -v
origin  https://github.com/virgogoo/python-check.git (fetch)
origin  https://github.com/virgogoo/python-check.git (push)
nguyet@ltn:~/Documents/my-openwrt-checkpy$
```